C5 - MULTIMODAL RECOGNITION

Task 3: Image Captioning (I)

Team 1:

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Setup & Introduction

Image Captioning

Process of generating **descriptive textual** sentences that convey the content of an image.

A person riding a motorcycle on a dirt road.

It integrates **Computer Vision (CV)** to automatically analyze visual data and **Natural Language Processing (NLP)** to produce coherent, contextually relevant captions that reflect the objects, actions, and scenes depicted in the images.

Goals

- **Primary**: Establish an Image Captioning baseline using an **Encoder-Decoder architecture** on the **VizWiz dataset** , evaluating performance with BLEU, ROUGE-L, and METEOR.
- **Modifications**: Test architectural changes by swapping the **Encoder** (e.g., ResNet, VGG) , **Decoder** (e.g., LSTM) , and **Text Representation** (subword, word-level, character).
- **Optional**: Implement an **attention mechanism**.

Frameworks & Models

- **Hugging Face (Evaluate Doc)**:
  - ROUGE-L, BLUE, METEOR
- **OpenAI (CLIP)**
- **Torchvision (Models Doc)**

torchvision

Hugging Face

OpenAI

- ***We first outline our experimental framework before detailing task results.***
- ***Github Repository: https://github.com/valy3124/C5-Team1***

Dataset: VizWiz

Note: These images were taken by blind persons. Because the photographers couldn't see the viewfinder, the images often suffer from poor lighting, extreme blur, "finger-over-the-lens," or the subject being entirely out of frame.

Model Development Phase

Train

Used for the initial training phase (**90% of the full training data**).

Training images: 20,579
Training captions: 90,432

Dev

Used for hyperparameter tuning and model selection (**10% of the full training data**).
Training images: 2,287
Training captions: 10,096

Final Training and Evaluation Phase

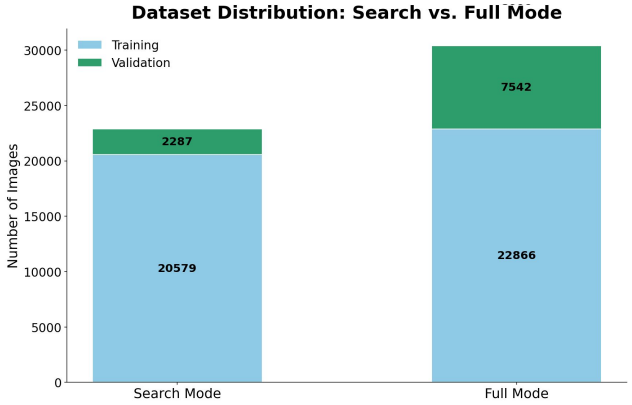
Full Train

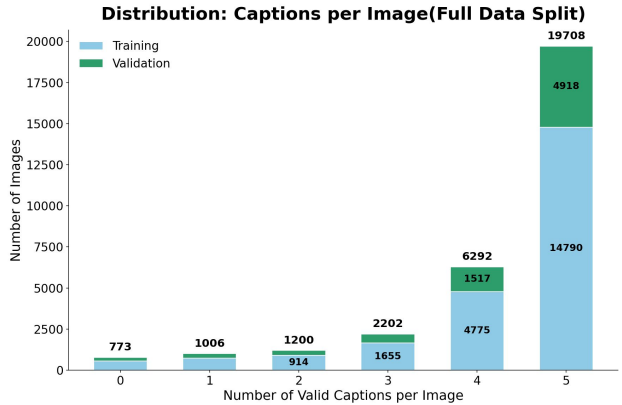
Used for the **final training stage**, combining Train and Train-Dev once the best model and hyperparameters are selected, to maximize available training data.

Testing

Reserved strictly for final evaluation. We used the official validation set.

Test images: 7,542
Test captions: 33,186





Key Observations

- There are some images with **no caption** that had to be removed.
- As per project requirement, we do **not use** the dataset's **test set**.
- There is a significant **imbalance** in the number of **captions per image**, which may affect model training.

Metrics Description

We use **METEOR** as our main metric during evaluation because it focuses on **recall** and handles **synonyms** (using a lexical database). This means it checks if the model actually gets the **meaning** of the image right, rather than just demanding exact word matches like BLEU does. We save **CIDEr** strictly for the final comparison between the baseline and our final model. Since CIDEr uses **TF-IDF** to give more weight to specific objects and less to common filler words, it's the most direct way to show whether our architecture changes actually **helped the model** recognize the important visual details in the scene.

METEOR *

Def: Harmonic mean of precision and recall (favoring recall), with a penalty for fragmentation.

$$F_{mean} \cdot (1 - \text{Pen})$$

where

$$F_{mean} = \frac{P \cdot R}{\alpha P + (1-\alpha) R},$$

$$\text{Pen} = \gamma \left(\frac{c}{m} \right)^{\beta}$$

Why: Rewards capturing the actual meaning and heavily penalizes scrambled word order.

BLEU

Def: Measures exact n-gram overlap with references (1 for words, 2 for pairs).

BLEU 1

$$\min\left(1, \frac{output\text{-}length}{reference\text{-}length}\right) precision_1$$

BLEU 2

$$\min\left(1, \frac{output\text{-}length}{reference\text{-}length}\right) \left(\prod_{i=1}^{2} precision_i\right)^{\frac{1}{2}}$$

Why: Verifies exact vocabulary match and penalizes artificially short sentences.

CIDEr

Def: Measures consensus with human references by weighting specific, informative words using TF-IDF.

$$\frac{1}{m} \sum_{j=1}^{m} \frac{\vec{\mathbf{g}}^n(c_i) \cdot \vec{\mathbf{g}}^n(s_{ij})}{\|\vec{\mathbf{g}}^n(c_i)\| \|\vec{\mathbf{g}}^n(s_{ij})\|}$$

where

- m: No. of human references.
- c_i: Candidate caption.
- s_{ij}: Reference caption j.
- $\vec{\mathbf{g}}^n$: TF-IDF weights.

Why: Ensures the model identifies key visual objects instead of just repeating common filler words.

ROUGE-L

Def: Measures the longest common subsequence (LCS) between the generated and reference text.

$$2 \cdot P \cdot RP + R$$

where

$$P = \frac{LCS}{c}, \; R = \frac{LCS}{r}$$

LCS: Length of the Longest Common Subsequence.
c,r: Total words in candidate (c) and reference (r).

Why: Ensures the correct order of visual events, even if words aren't perfectly consecutive.

* Michael Denkowski, Alon Lavie(2014) Meteor Universal: Language Specific Translation Evaluation for Any Target Language. In ACL Workshop on Statistical Machine Translation

Approach: Experimental Strategy

Core Evaluation Premises

- **Primary Metric**: **METEOR** drives all decisions. All metrics are obtained with the models achieving best METEOR in the validation set during training.
- **Data Integrity**: Steps 0–5 use strictly **Train/Dev** splits, preventing data leakage.

Train only the **Baseline** and the **Final Model** on the **Full Training Set**. Evaluate on the **Test Set** (Slide 3) to compare generalization and perform qualitative analysis.

Step 0: Baseline

- Train **initial architecture** (**ResNet-18 + GRU + Char-level**).
- Set baseline metrics and establish data evaluation methods.

Step 1: Encoder Search

- **Fixed**: Decoder & Char-level tokenization.
- **Action**: Explore various vision encoders and select the best.

Step 2: Decoder Search

- **Fixed**: Best Encoder & Char-level tokenization.
- **Action**: Explore decoder architectures (**GRU**, **LSTM**, **xLSTM**), layer depths, and embedding dimensions. Select the best configuration per type.

Step 3: Text representation

- **Fixed**: Best Encoder & Decoders.
- **Action**: Evaluate tokenization levels (**Character**, **Subword**, **Word**). Select the best performer.

Step 4: Attention Mechanisms

- **Fixed**: Best Encoder, Decoder, & Word-level tokenization
- **Action**: Explore attention types (**Soft**, **Adaptive**, **Visual Prefix**) to improve visual grounding.

Step 5: Hyperparameter search

- **Fixed**: Best overall architecture
- **Action**: Perform a Bayesian sweep (**Optimizer**, **LR**, **Scheduler**, **Decoding method**) to find the optimal final model

STEP 0: Baseline

Setup

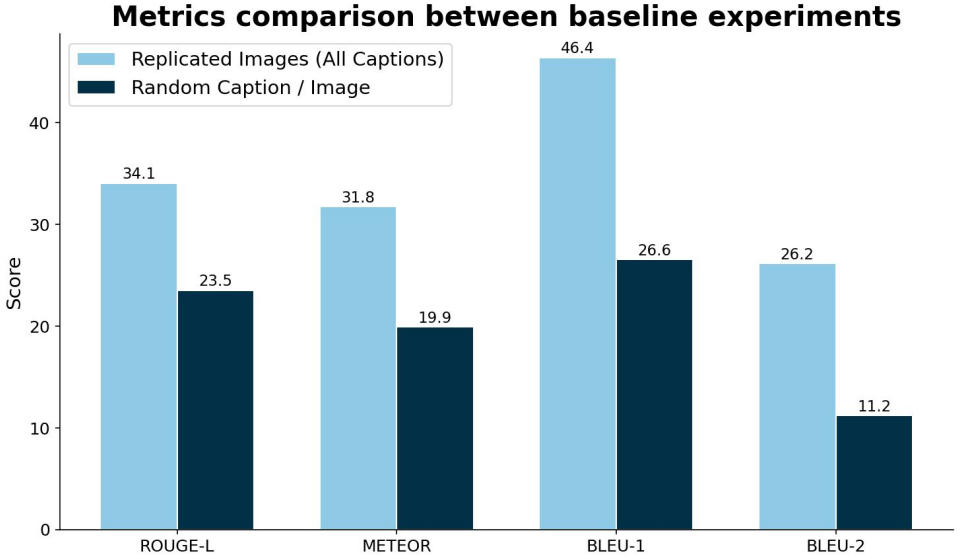
- **Baseline Model:** ResNet18 + GRU + Characters
  Optimizer: Adam | **LR:** 5e-5
  Scheduler: Plateau | **Batch size:** 128
- **Data Strategies Tested:**
  - **Random Caption**: Selects one random caption per image per epoch for both training and validation.
  - **Replicated Images (All Captions)**:
    - **Duplicates images** so each available caption forms a unique training pair, effectively **multiplying our training samples**.
    - During evaluation, the model's prediction is compared against **all ground-truth references** for that image, returning the **highest matching score.**

Decision

- **Final Setup:** We will use the **Replicated Images** strategy with multi-reference evaluation for all subsequent experiments in this project.

Results

- The "Replicated Images" strategy **significantly outperforms** the "Random Caption" method across all metrics (e.g., METEOR jumps from 19.9 to 31.8).
- **Why the improvement?** Training on all available captions exposes the model to a **larger volume of data** and more **diverse sentence structures** for the same visual features. Furthermore, evaluating against all references is a **much fairer reflection of performance**, as a single image can have multiple perfectly valid descriptions.



STEP 1: Encoder Search

Setup

- **Baseline:** Resnet18 *(METEOR: 31.8)*
- **Setup:** Encoder freezed for all the experiments.

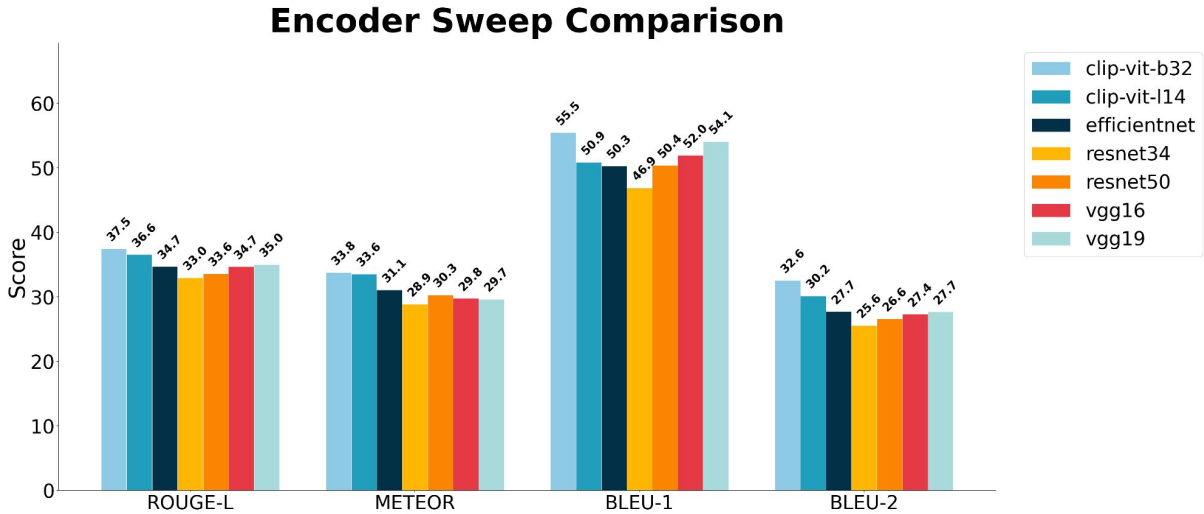
- **Encoder sweep:**
  - **CLIP-ViT-b32** (89.5M params)
  - **ResNet-34** (22.9M params)
  - **VGG-16** (16.4M params)
  - **Efficientnet-B0** (6.3M params)
  - **ResNet-50** (26.2M params)
  - **VGG-19** (21.7M params)
  - **CLIP-ViT-L14** (305M params)

Results

- **CLIP-ViT-b32** achieves the highest scores across all metrics, outperforming all other encoder backbones.
- CLIP is pre-trained on **image–text pairs**, so the obtained features are already **semantically aligned**, making them more useful for the captioning decoder than CNN architectures.
- **Model size ≠ performance:** Larger encoders don't consistently improve results; representation quality matters more than parameter count.

Decision

- **Final model:** CLIP-ViT-b32
- **METEOR:** 33.8
- **Improvement** +2.0



STEP 2: Decoder Search

Setup

- **Baseline:** GRU (1 layer, 512 dim) *(METEOR: 33.8)*
- **Setup:** Encoder frozen; search best hidden dim and number of layers per decoder type.
- **Decoder sweep**
  - **Types:** GRU, LSTM, xLSTM
  - **Hidden dim**: 512, 1024
  - **Layers**: 1, 2, 3

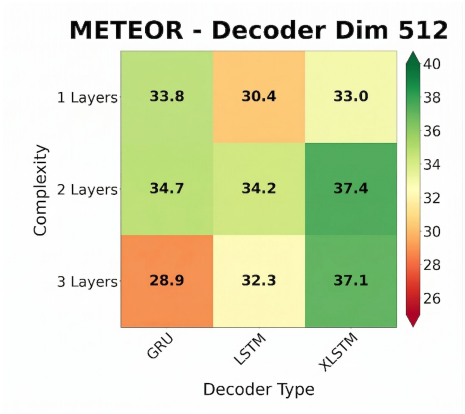
Results

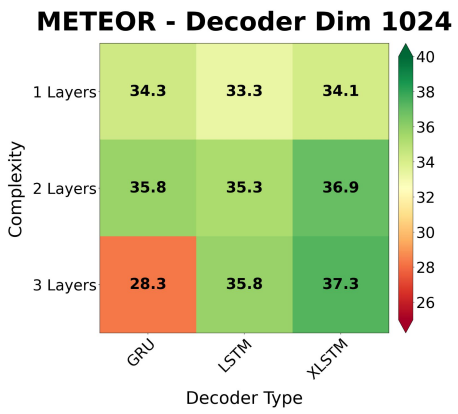
- **Best decoder:** *xLSTM* consistently achieves the top performance across configurations, with a small but stable margin over LSTM.
- **Depth vs stability:** Performance peaks at **2 layers**, with deeper GRU models degrading sharply, whereas xLSTM maintains stable performance even at higher depth.
- **Width vs architecture:** Increasing hidden size benefits LSTM, but **cannot compensate for weaker architectures** (GRU). Overall, decoder type has a larger impact than dimension size

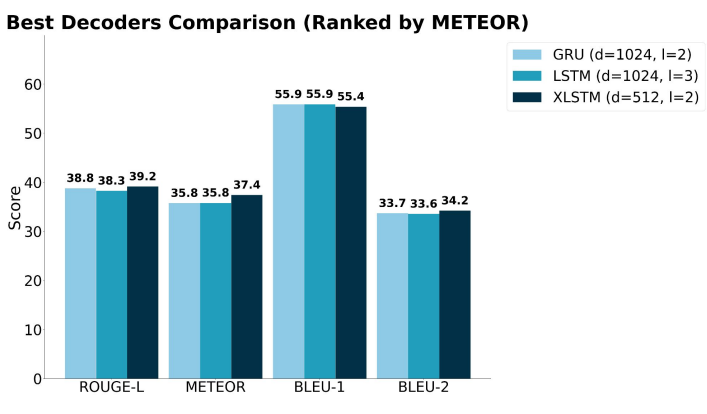
Decision

Best configuration per decoder

- GRU: 2 layers, 1024 dim
  - **METEOR**: 35.8 (+2.0)
- LSTM: 3 layers, 1024 dim
  - **METEOR**: 35.8 (+2.0)
- xLSTM: 2 layers, 512 dim
  - **METEOR**: 37.4 (+3.6)







STEP 3: Text Representation Search

Setup

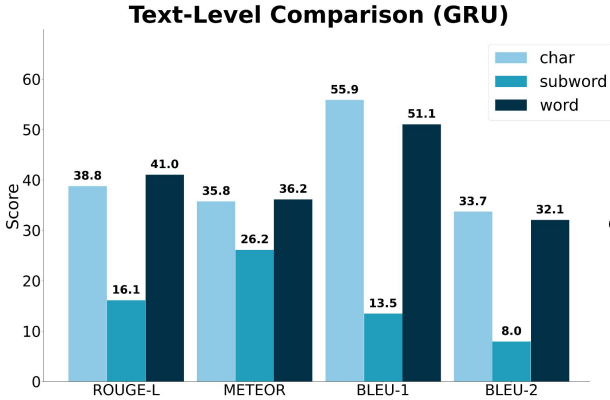
- **Baseline** (character tokenization):
  - **GRU:** METEOR **35.8**
  - **LSTM**: METEOR **35.8**
  - **xLSTM**: METEOR **37.4**
- **Setup:** Compare tokenization strategies across decoders
- **Tokenization sweep: Character**, **Subword** and **Word**

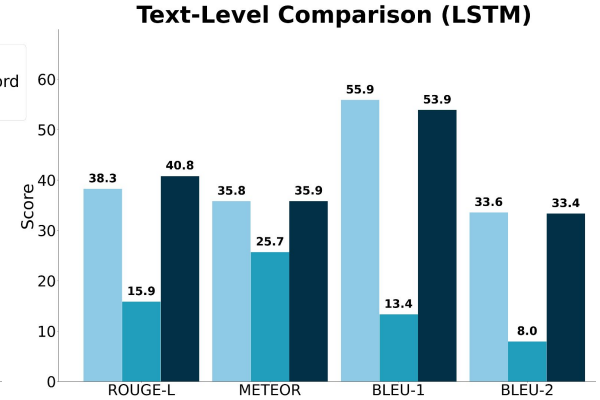
Results

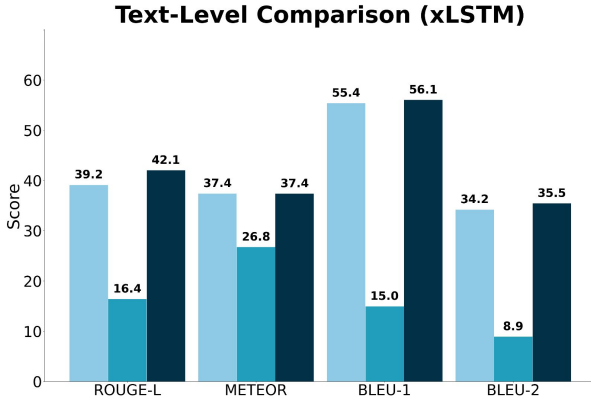
- **Subword collapse:** Subword tokenization performs significantly worse, as models struggle to **properly terminate sequences**, often looping and repeating fragments instead of forming complete words.
- **Word vs character:** Both perform strongly. Word-level provides **direct semantic tokens**, while character-level uses a **closed vocabulary**, avoiding "out-of-vocabulary" issues and enabling flexible word construction.
- **Consistent trend:** The same ranking holds across **GRU**, **LSTM**, and **xLSTM.** Tokenization choice has a larger impact than decoder architecture.

Decision

- We select **word-level tokenization** for attention across all decoders (better semantic alignment)
  - GRU**: METEOR**: 36.2 (**+0.4**)
  - LSTM**: METEOR**: 35.9 (**+0.1**)
  - xLSTM: **METEOR**: 37.4 (**+0.0**)







STEP 3: Qualitative text-representation

VizWiz_train_00011300.jpg

GT Captions

- **An analog clock is captured on a wooden desk.**
- **A cable box for the TV showing the time.**
- **A gadget with a timer on the kitchen counter top**
- **A radio is on the bedside table showing time.**

We chose a validation set sample to demonstrate the qualitative differences in different text-representation levels using a LSTM decoder.

LSTM - Char level

A black electronic device with a silver and black button and a silver background and a silver background and a silver background with a blue background and white text on it.

LSTM - Subword level

a digital clock on a table with a digital clock . 0 0 . 8 on it . 2 012 . 11 . 0 0 degrees . 01. 0 80 /2012 . 1 2 degrees . 1 3 . 8 degrees fahrenheit . 2 012. 80/ 22. 80/ 22 pm

LSTM - Word level

a digital clock with the time <UNK> and the time <UNK>

Observations

- **Char-level repetition**: It completely misses the main object (the clock) and gets stuck in a loop repeating colors and backgrounds.
- **Subword-level hallucination**: We see that it correctly identifies the "digital clock," but it wildly invents random dates, temperatures, and numbers to make up for the blurry text.
- **Word-level handles uncertainty with <UNK>**: It provides the most logical sentence structure but outputs <UNK> (Unknown) tokens when it cannot deal with out-of-vocabulary words, like the numbers on the display.

* ***Word*** *and* ***Char-level*** *representations perform better than* ***Subword-level*** *but remain imperfect. Next, we will see new modules that solve these remaining issues.*

STEP 4: Soft Attention (*)

Note: Attention mechanism used in **LSTM** and **GRU**.

Motivation: Traditional image captioning compresses the entire image into a single global vector, losing spatial details. **Soft Attention** solves this by computing a differentiable, weighted distribution over all image patches. Unlike **Hard Attention**, which rigidly selects only one single patch at a time and requires complex reinforcement learning to train. Soft Attention focuses its "gaze" on relevant regions while remaining fully trainable via backpropagation.

Step 1: Extract Annotation Vectors

- Pass the image through a encoder to **extract L spatial patches**. Each patch is a vector representing a specific region of the image.
- This will be our **annotation vectors** (one for each patch):

$$a = \{a_1, \ldots, a_L\}$$

Step 2: Calculate Attention Weights (Bahdanau Additive Attention)

- At time step t, the decoder's previous hidden state h_(t-1) **evaluates each patch a_i to calculate an alignment score**, which is then converted into a probability distribution (**weights**) that sums up to 1.

$$e_{i,t} = v_{att}^T \tanh(W_a a_i + W_h h_{t-1})$$

$$\alpha_{i,t} = \frac{\exp(e_{i,t})}{\sum_{j=1}^{L} \exp(e_{j,t})}$$

Step 3: Compute the Context Vector

- **Multiply each image patch by its corresponding attention weight and sum the results**. This creates a single dynamic vector **focused only on the most relevant visual features for the current word**.

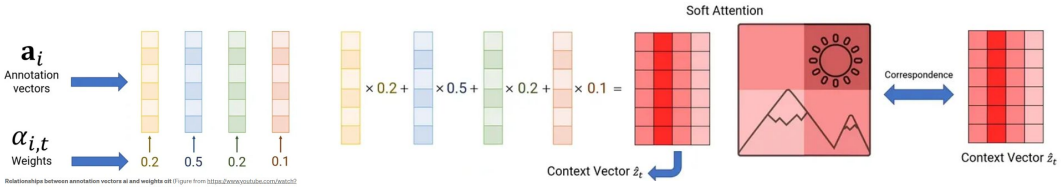
$$\hat{z}_t = \sum_{i=1}^{L} \alpha_{i,t} a_i$$

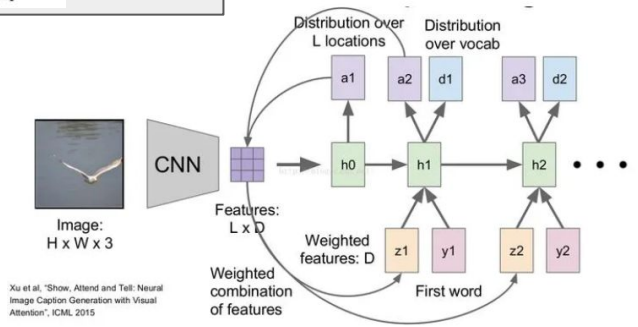
Step 4: Update Decoder and Generate

- Feed this visual context vector alongside the previous word embedding into the decoder cell to **update its memory state and predict the next word.**

$$h_t = \text{Decoder}([Ey_{t-1}; \hat{z}_t], h_{t-1})$$

Example for 4 Patches





* K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. Zemel, and Y. Bengio, "Show, attend and tell: Neural image caption generation with visual attention," in *International Conference on Machine Learning (ICML)*, 2015, pp. 2048-2057.

STEP 4: Adaptive Attention (*)

Note: Attention mechanism used in **LSTM** and **GRU**.

Motivation: Standard attention mechanisms force the decoder to look at the image for *every* generated word. However, non-visual words (like "the" or "of") and **highly predictable context do not require visual grounding**. **Adaptive Attention** introduces a ***Visual Sentinel***, allowing the model to dynamically decide at each time step whether to attend to the image or rely solely on its internal language memory.

Step 1: Generating Visual Sentinel (s_t)

- The sentinel represents the **linguistic knowledge** of the decoder. It acts as a buffer that stores what the model has already learned from the text sequence so far.
- A **gating mechanism** (g_t) decides **how much** of the current RNN state should be saved as a visual fallback.

$$g_t = \sigma(W_x x_t + W_h h_{t-1}) \quad s_t = g_t \odot f(\text{internal\_state}_t)$$

Step 2: Adaptive Attention Scores

- We calculate how well the current state matches the image regions **versus** how much it matches the sentinel.
- **Image Scores z_(t,i)**: Measures relevance of visual region i.
- **Sentinel Score z_(t,s)**: Measures the desire to ignore the image and rely on the language model.

$$z_{t,s} = w^T \tanh(W_s s_t + W_g h_t) \quad z_{t,i} = w^T \tanh(W_v v_i + W_g h_t)$$

Step 3: The Sentinel Gate (β_t)

- This is the Decision Step. We use a Softmax to create a probability distribution **over the image AND the sentinel.**

$$\hat{\alpha}_t = \text{softmax}([z_t; z_{t,s}]) \quad \hat{\alpha}_t = [\underbrace{\alpha_{t,1}, \alpha_{t,2}, \ldots, \alpha_{t,k}}_{\text{Spatial Attention } (\alpha)}, \underbrace{\beta_t}_{\text{Sentinel Gate}}]$$

- The **last element k+1 is the Sentinel Gate β_t**: the specific weight given to the sentinel.

Step 4: Building the Adaptive Context Vector and Word prediction

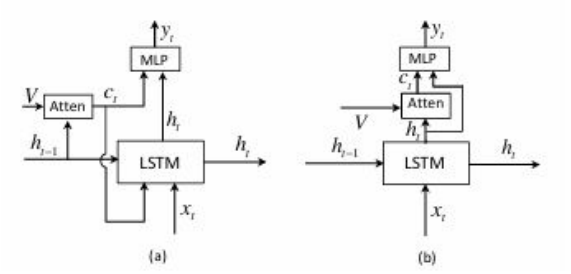
- The final context vector is a **mixture of the visual features and the sentinel**, regulated by the weight β_t.

$$\hat{c}_t = \beta_t s_t + (1 - \beta_t) c_t$$

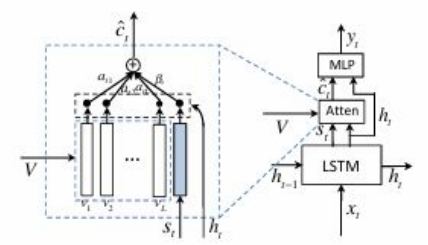
- The model looks at **both the Adaptive Context and its own Hidden State** to pick the next word.

$$P(y_t) = \text{softmax}(W_p(\hat{c}_t + h_t))$$

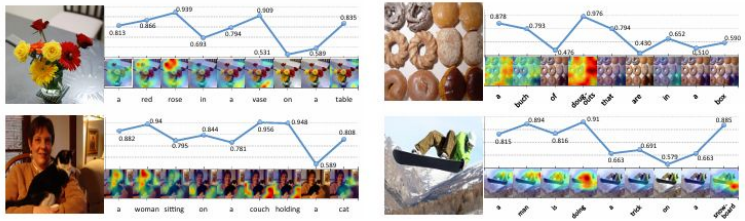
Soft Attention (a) vs Adaptive Attention (b)



Adaptive Attention (more detail)



Visual Grounding probabilities of each generated word and corresponding spatial attention maps



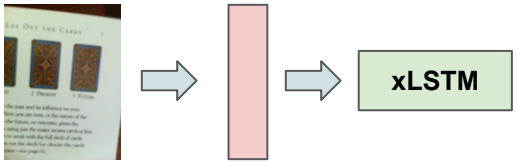
* J. Lu, C. Xiong, D. Parikh, and R. Socher, "Knowing when to look: Adaptive attention via a visual sentinel for image captioning," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 3242-3250.

STEP 4: “Visual Prefix” in xLSTM

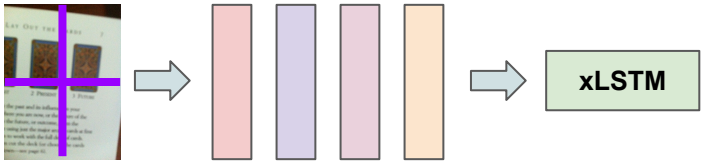
Note: Only used in **xLSTM.** This is **not an attention mechanism**.

Motivation: Traditional models suffer from a spatial information bottleneck by compressing the entire image into a single vector. To avoid the high computational cost of step-by-step explicit attention, here we use a **Visual Prefix**. By sequentially injecting image patches before the text, xLSTM's novel **Matrix Memory** natively absorbs and retains complex spatial relationships.

Default Strategy



Visual Prefix



Explanation

- **Sequential Visual Prompting:** Instead of applying global pooling, the uncompressed spatial feature map is flattened into a sequence of N patches. These are concatenated with a learnable **<SEP>** token to form a unified multimodal input stream: **[Patch_1, ..., Patch_N, <SEP>, Word_1, ...].**
- **Covariance Matrix Conditioning:** As xLSTM sequentially processes the visual prefix, its internal covariance memory matrix C_t iteratively absorbs the visual patches. This builds a rich, structural representation of the image space within the recurrent state.
- **O(T) Inference Efficiency:** The decoder auto-regressively predicts words **by querying this heavily conditioned matrix memory**. This allows the model to ground words to spatial regions **without computing expensive** explicit attention weights at every step, reducing complexity to O(T).

STEP 4: Attention Quantitative

Setup

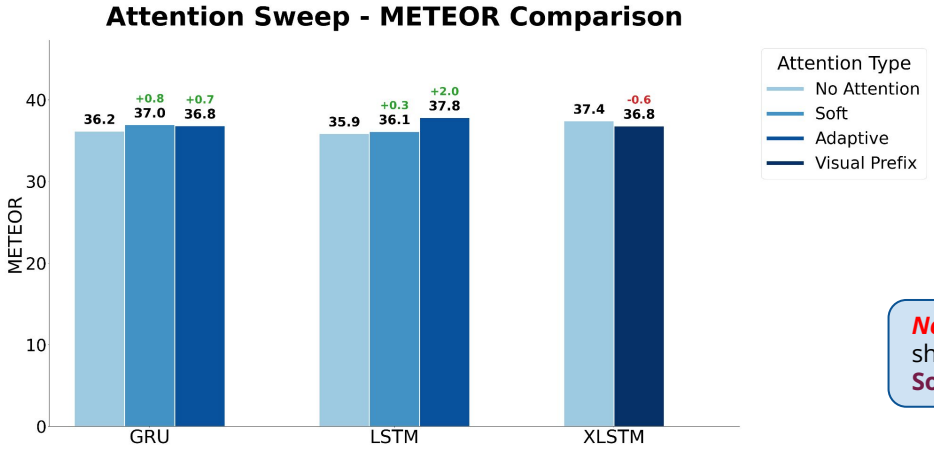
- We explored **two attention mechanisms** for **GRU** and **LSTM** decoders: **Soft Attention** and **Adaptive Attention**.
- For **xLSTM**, we adopted a **Visual Prefix** approach, representing images as sequences of visual tokens instead of using attention.

Results

- Attention **consistently improves performance over the baseline** for **GRU** and **LSTM**. For **GRU**, both attention mechanisms yield **similar gains**, while for **LSTM**, **Adaptive Attention** significantly **outperforms Soft Attention**, likely due to its ability to **dynamically balance visual information and language context via the sentinel mechanism.**
- For **xLSTM**, **Visual Prefix** **does not improve performance**. This is likely because **xLSTM** already integrates long-range dependencies through its internal memory, and **lacks explicit mechanisms to re-attend to spatial features**, making it harder to refine visual grounding during generation.

Decision

- **Final Model: LSTM + Adaptive Attention**
- **METEOR: 37.8**
- **Improvement: +0.4** (with respect to the best metric in the previous experiment)

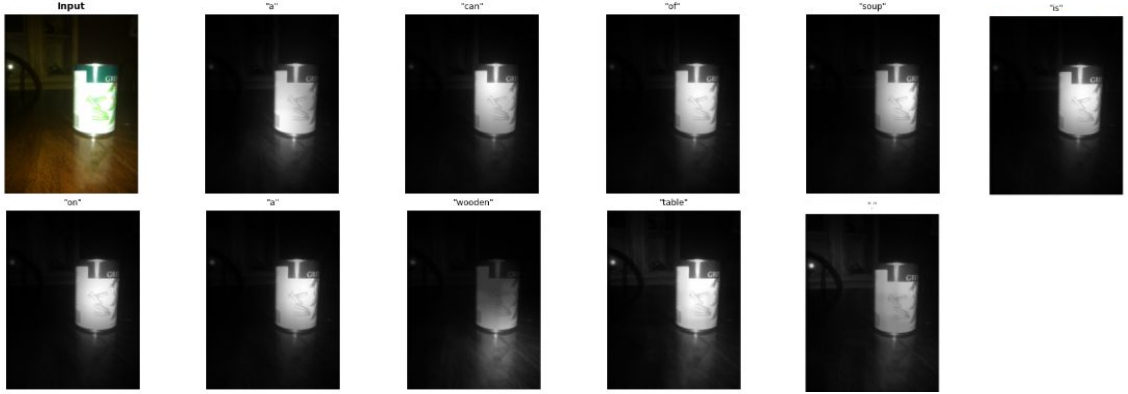


Note: In the following slides, we show qualitative examples for both **Soft** and **Adaptive Attention**.

STEP 4: Soft Attention - Qualitative Examples

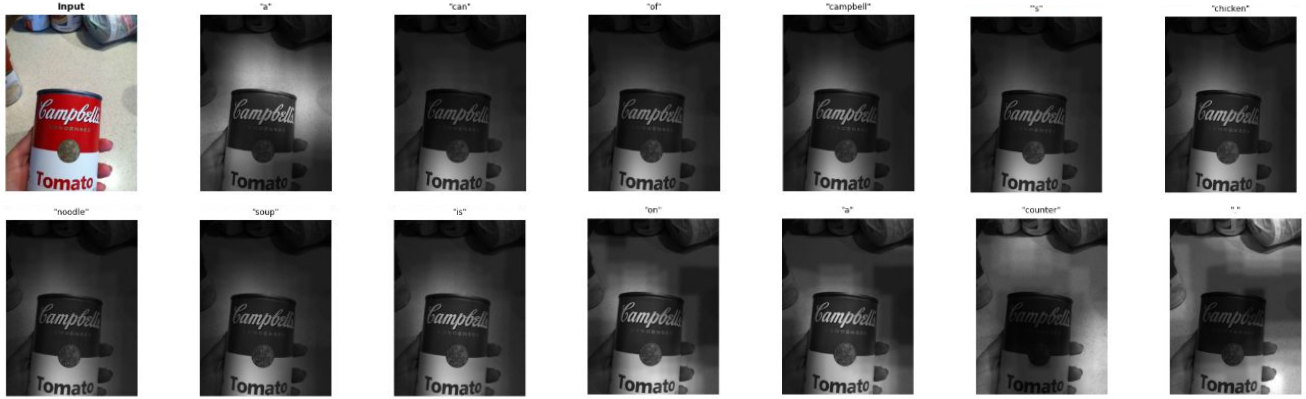
Ex 1

Pred: a can of soup on a wooden table



Ex 2

Pred: a can of campbell's chicken noodle soup on a counter

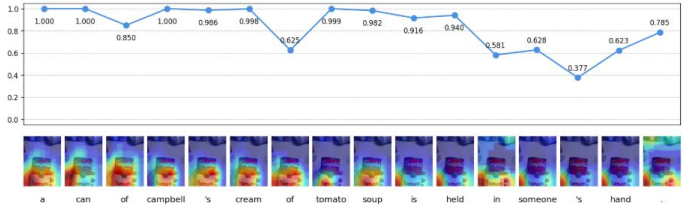


Key Observations

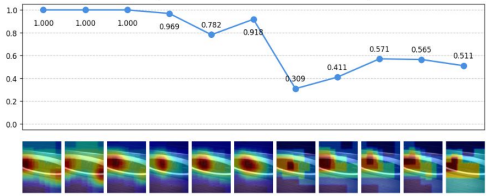
- The model is **forced to attend to the image at every timestep**, ensuring continuous visual grounding (we have based on the **visualizations of the original paper**).
- Attention maps are stable and consistently focused on the main object, with **minimal drift over time**. Some variations can be observed **in Example 1** for words such as **"wooden"** and **"table"**, where attention **shifts towards relevant contextual regions.**
- **It provides better results than the baselines but not better than adaptive attention.**

STEP 4: Adaptive Attention - Qualitative Examples

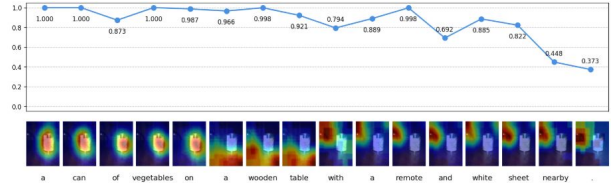
Ex 1



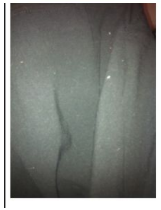
Ex 2

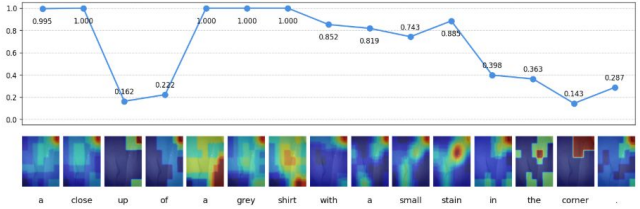


Ex 3



Ex 4





Key Observations

- The line chart represents **1−β_t,** indicating how much the model relies on **visual information vs. language context** at each timestep.
- **Example 1:** Function words (e.g., “of”, “’s”) show lower values, which is expected since **they rely more on language modeling**. In contrast, content words exhibit meaningful attention maps:
  - **“Campbell”**: focus on the center of the can.
  - **“Tomato”**: focus on the lower region.
- **Examples 2 and 3:** The model behaves as expected: high values for key visual tokens attention maps remain semantically aligned with relevant regions.
- **Example 4:** Lower and unstable values. Attention maps appear diffuse and blocky, likely due to low structure / ambiguous visual content, making it harder to extract meaningful features.

With this method we have got our best results.

STEP 5: Hyperparameter Search *(Beam Search)*

Motivation: Before the final hyperparameter sweep, we introduce **beam search** as it directly affects caption generation quality. Unlike greedy decoding, beam search explores multiple candidate sequences, leading to more accurate and fluent captions.

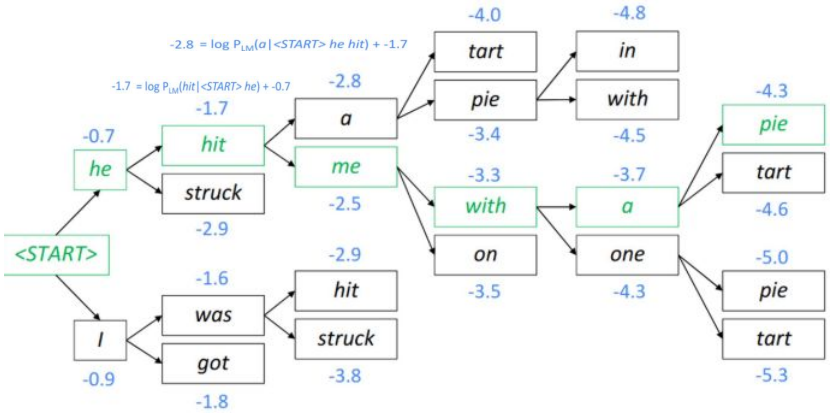
How does it Work?

- At each timestep, keep the **top K sequences (beams).**
- Expand each beam with all possible next words.
- Keep the **K best sequences based on score.**
- **For each beam:**

$$score = score_{prev} + \log P(next\_word)$$

$$\log P(y_{1:T}) = \sum_{t=1}^{T} \log P(y_t \mid y_{1:t-1}, x)$$

Beam Search with K = 2



For each of the hypothesis, find top K next words and calculate scores.

Comparison with Greedy Decoding

Differences:

- **Greedy decoding** is **locally optimal** (selects the most probable word at each step). **Beam search** is **globally optimized** (approx) (keeps multiple candidates and selects the best overall).

Advantages of Beam Search:

- **Produces more coherent and fluent captions.**
- **Better global sequence optimization.**

Disadvantages of Beam Search:

- **Higher computational cost (slower inference).**
- **Performance depends on beam size (K).**

STEP 5: Hyperparameter Search

Note 1: SGD had the strongest negative correlation and is omitted in the importance plot to preserve plot clarity.
Note 2: We performed limited runs due to high computational cost (~3.5h per experiment).

Setup

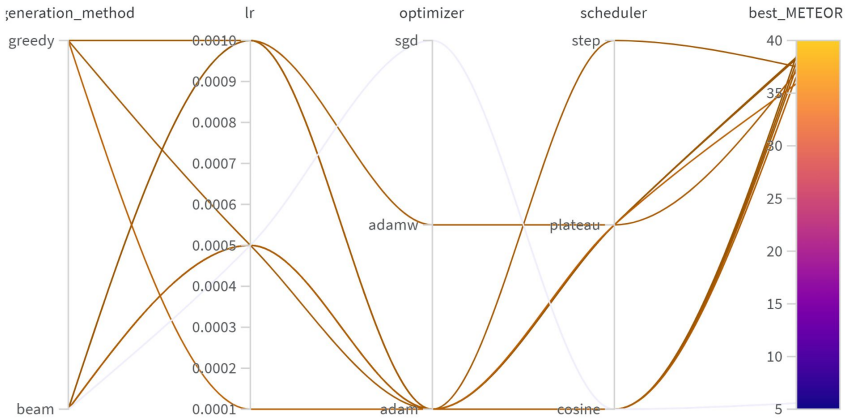
- Using the best configuration so far **(CLIP-ViT-b32 + LSTM + Word + Adaptive Attention)**, we perform a bayesian hyperparameter sweep over the following settings:
  - **Optimizer**: **SGD**, **AdamW**, **Adam**
  - **Learning rate**: **1e-3**, **5e-4**, **1e-4**
  - **Scheduler**: **Step**, **Plateau**, **Cosine**
  - **Decoding**: **Greedy**, **Beam Search (K=3)**

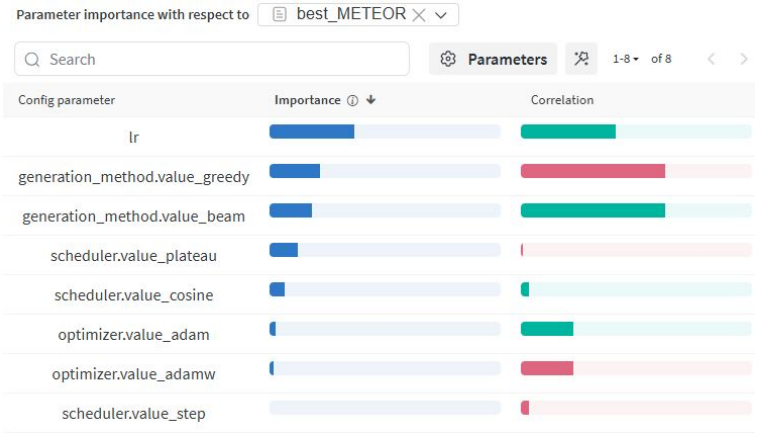
Results

- **Decoding:** Beam search outperforms greedy by exploring multiple sequences, avoiding early suboptimal choices and improving coherence.
- **Optimizer:** SGD underperforms due to lack of adaptive learning rates, making it slower and less effective in complex, noisy optimization like sequence generation.
- **Learning rate:** Most impactful hyperparameter. Lower values (**1e-4**) lead to more stable training and better performance.

Decision

- **Final Model:**
  - **Optimizer: Adam**
  - **Learning rate: 1e-4**
  - **Scheduler: Cosine**
  - **Decoder: Beam Search**
- **METEOR: 38.56**
- **Improvement: +0.7**





Final Models: Generalization

Setup

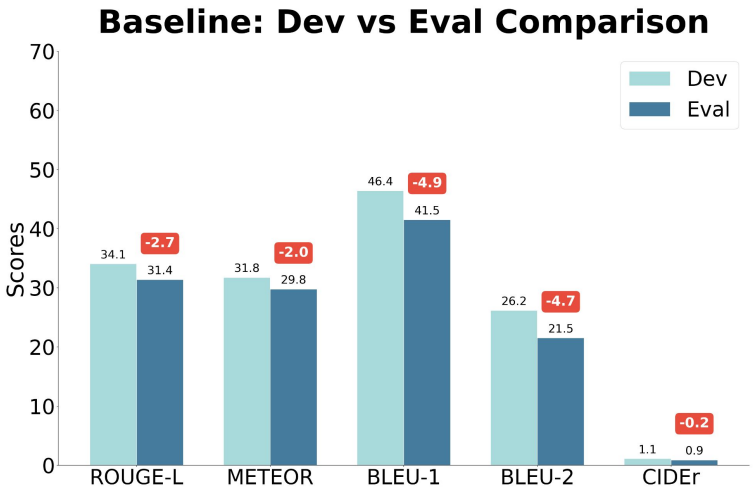
- We compare the **baseline** and our **final model** across two different data splits (slide 3) to assess the **generalization gap.**
- **Baseline Model:** ResNet18 + GRU + Characters
  Optimizer: Adam | **LR:** 5e-5
  Scheduler: Plateau | **Decoding:** Greedy
- **Final Model:** CLIP_ViT_b32 + LSTM + Word + Adaptive Attention
  Optimizer: Adam | **LR:** 1e-4
  Scheduler: Cosine | **Decoding:** Beam

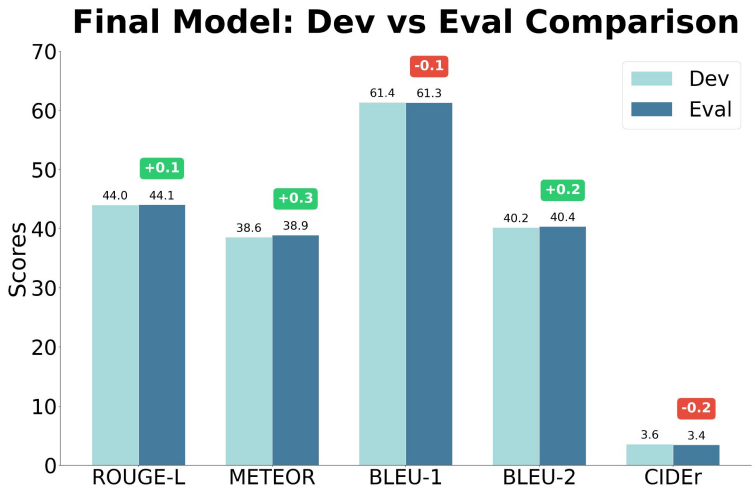
Results

- The final model shows **clear improvements** over the baseline across both data splits (Dev and Eval), with **higher scores in all the evaluated metrics**.
- Regarding generalization, the gap between Dev and Eval performance in the final model is **slightly positive** (+0.3) in METEOR, our main metric, which suggests that the model generalizes well and does not suffer from overfitting.

Comparison

- **Improvement in Dev: +6.8**
- **Improvement in Eval: +9.1**
- **Generalization Gap for Final Model: +0.3**





Qualitative per-step Comparison

Setup

- **Objective:** To visualize qualitative improvements across architectural upgrades.
- **Method**: Comparing predictions from the best epoch of each major model iteration with the ground truth captions.
- **Pipeline**: **Baseline** ➔ +CLIP **Encoder** ➔ +LSTM **Decoder** ➔ +**Word-level tokens** ➔ +**Attention** ➔ **Best Sweep**.

VizWiz_train_00006514.jpg

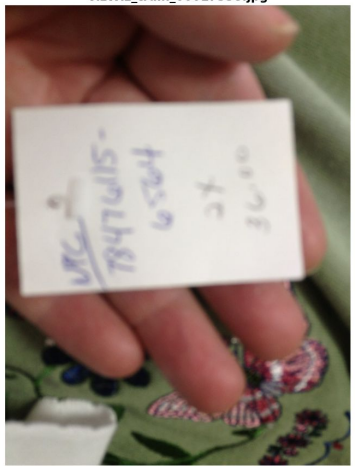
Ground Truths:

1. A package of Butterball brand turkey franks hot dogs.
2. blue and beige colored bag of turkey hot dogs on a beige countertop
3. An unopened package of Butterball Turkey Franks resting on a grainy surface.
4. A package of turkey hot dogs that are lactose free.
5. A white hot dog package laying on a counter top

Model Predictions:

Baseline (Ep 9):	A package of food is on top of a table.
Encoder (Ep 6):	A package of frozen pasta sauce with a picture of the food in it.
Decoder (Ep 9):	A blue bag of frozen spinach with a blue and yellow label.
Text-level (Ep 9):	a package of chicken breast fillets is on a table .
Attention (Ep 7):	a blue and yellow package of jimmy dean hot dogs laying on a gray surface .
Sweep (Ep 7):	a blue and yellow package of butterball brand frozen food .

VizWiz_train_00017338.jpg



Ground Truths:

1. A white label has text written in pen
2. a human hold sticker on his hand with a text closely related to phone number
3. A hand holding a piece of paper with writing from a pen on it.

Model Predictions:

Baseline (Ep 9):	A hand holding a blue box with a barcode on it.
Encoder (Ep 6):	A person holding a pink shirt with a pink shirt with a black and white label.
Decoder (Ep 9):	A person is holding a piece of paper with a picture of a person in the background.
Text-level (Ep 9):	a person holding a piece of paper with a signature on it .
Attention (Ep 7):	a person holding a small white box with a handwritten number on it
Sweep (Ep 7):	a hand is holding a white piece of paper .

Conclusions

- **Final Sweep:** It consistently provides the most **logical**, detailed, and accurate descriptions of the images.
- **Early models hallucinate**: Baseline and early epoch models struggle heavily, frequently **inventing objects** not present in the scene.
- **Text recognition improves over time:** Later models successfully **detect and read text** in the wild(as the brand name).

Qualitative Comparison: *Baseline VS Final Model*

Context: We conducted a qualitative comparison of the **final baseline** (best Epoch 10) and **final model** (best Epoch 7), both trained on the **full training set** and selected for their optimal METEOR scores on the validation set.

VizWiz_val_00000052

GT Captions

- **A white label has a barcode on it and a Westell company logo.**
- **A manufacturing label with a logo and barcode.**
- **WHITE COLORED RECTANGULAR SHAPED PRINTED SENTENCE PLACED ON FLOOR.**
- **a close up of a sticker of a manufacture information of a type of electronic device, that says "Westell".**

Final Baseline Prediction

A package of food is on top of a table.

Final Model Prediction

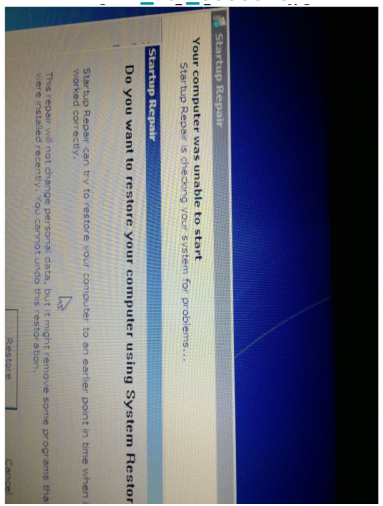
a label with a barcode and a barcode on a black surface

Conclusions

- The final model **clearly improves over the baseline**, producing a caption that better matches the ground truth by correctly identifying key elements like the **label and barcode**, whereas the baseline is semantically incorrect.
- However, the prediction still contains **redundancies** (e.g., “a barcode and a barcode”) and remains a bit generic.

Qualitative Evolution: *Final model*

VizWiz_val_00000404



GT Captions

- **A screen is turned on with a startup repair screen.**
- **Sideways view of the Startup Repair window on a computer.**
- **Pictured is a computer screen turned sideways in startup repair mode.**
- **A computer screen part blue with a pop up screen on it about repairing the computer.**

Epoch 1
a computer screen showing a computer screen with a message.

Epoch 2
a computer screen showing a startup repair window.

Epoch 3
a computer screen with a pop up message.

Epoch 4
a computer screen with a pop up message on it.

Epoch 5
a computer screen is displaying a window with a message

Epoch 6 - BEST
a startup repair dialogue box on a computer.

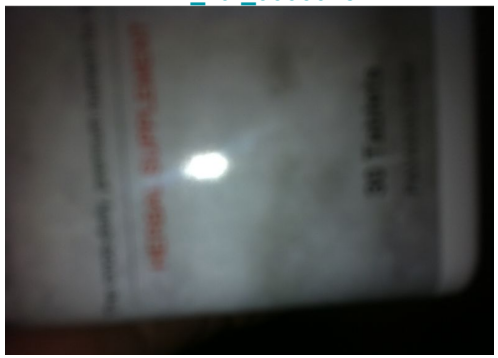
Epoch 7
a pop up message on a computer screen .

Conclusions

- **Contextual Understanding**: Comparing between epochs, the best one in terms of METEOR **successfully reads** the specific context of the image, instead of generating the generic description “pop up message”.
- **Learning is non-linear**: We can see how the model fluctuates during training. It starts with repetitive guesses, defaults to generic answers in the middle and finally **aligns with the ground truth** at its peak.
- **Epoch 2 vs Epoch 6**: Both capture the correct semantic meaning, but Epoch 6 **refines** from “windows” to “dialogue box”.

Qualitative Examples: *Final Model*

VizWiz_val_00000254



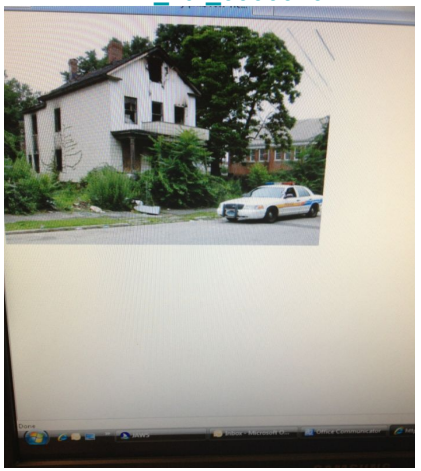
Caption - BAD

an advertisement for an iphone is being held here

GT Captions

- **A white bottle of something with writing on it.**
- **looks like a board in the dark with a light shining on it.**

VizWiz_val_00000102



Caption - GOOD

a computer screen with a picture of a car on it

GT Captions

- **A house with some police car in the monitor window.**
- **Computer display monitor displaying old home and police vehicle.**
- **Computer screen display of a picture of a fire damaged house with a police car parked in front of it.**
- **A computer screen with a photo of an abandoned White House with an overgrown garden and a police car parked in front of it.**

Conclusions

- The overall quality of the dataset is **limited**, which sometimes **leads the model to generate suboptimal or forced captions**, negatively affecting generalization on the evaluation set (as illustrated in the example on the left).
- For the high quality images, the model is **able to produce nice captions** like in the example on the right

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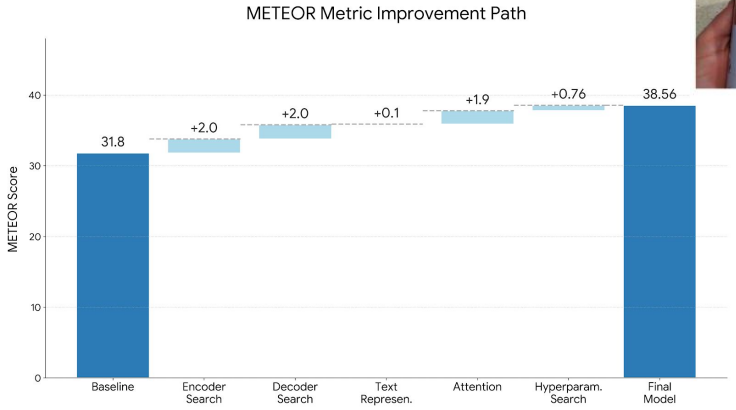
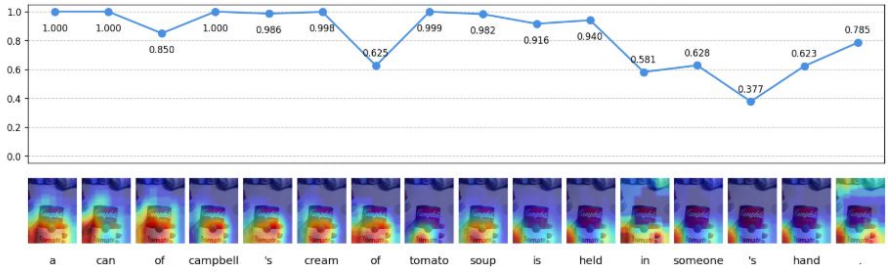
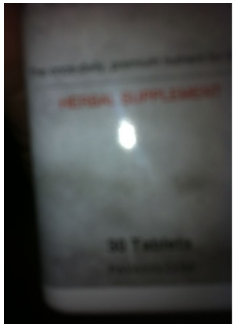
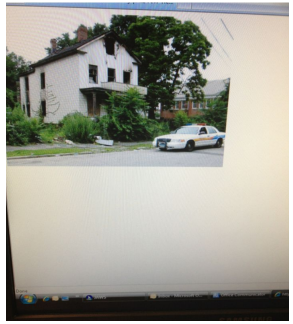
Summary







an advertisement for an iphone is being held here



a computer screen with a picture of a car on it

Conclusions

- **Tackling a Highly Irregular Dataset**: The VizWiz dataset presents severe challenges, featuring **images with poor lighting, extreme blur, or fingers over the lens**, paired with equally strange and noisy GT captions that heavily test the model's limits.
- **Pre-trained Semantic Alignment Wins**: The **CLIP encoder outperformed traditional CNNs** (ResNet/VGG) because its features are already aligned with text, making the decoder's job much easier.
- **Architecture vs. Attention Dynamics**: Without explicit attention, xLSTM outperforms both LSTM and GRU. However, when paired with Adaptive Attention, **LSTM ultimately surpasses xLSTM's performance**.
- **Vocabulary Granularity Matters**: **Word-level tokenization proved to be the most robust**. Subword levels struggled with sequence termination, leading to repetitive loops and severe hallucinations.
- **Knowing "When" to Look is Crucial**: **Adaptive Attention was the most effective mechanism**. By using a "visual sentinel," the model successfully learned when to ground words in the image and when to rely on its internal language model (e.g., for syntax and grammar).
- **Overall Evolution**: Through systematic upgrades (CLIP + LSTM + Word-level + Adaptive Attention), we evolved the model from outputting repetitive hallucinations to **successfully reading natural scene text** and accurately describing complex scenes.